# Where does weight-space programming fix failures?

NVIDIA Cosmos3 Edge Policy (world-action model) and NVIDIA GR00T N1.7 DROID (VLA on a Cosmos-Reason2-2B backbone), with OpenPI π0.5 as the control, on the same RoboLab DROID tasks T1–T8 and a retention task R1.

**Method.** When a clearer wording of the same goal makes a robot model succeed, copy that wording's internal effect into the weights in closed form, one least-squares solve per layer: no training, and the instruction stays the same.

**π0.5, the control (earlier records).** Distillation fixed its two language failures: T1 27 → 106 of 128 unseen starts (81 fixed, 2 broke, $p = 7.2 \times 10^{-22}$) and T8 0 → 27 of 64; a random edit of the same size did nothing (9 vs 13 of 64); the T1 edit also lifted T2 and T3; no edit helped T4–T7.

**NVIDIA models.** Cosmos3 Edge Policy: unedited rates in, edits running; GR00T N1.7: unedited rates in, edits running. First finding: GR00T N1.7 succeeded on 5 of 16 starts under RoboLab's performance render preset and 16 of 16 under the default preset, so all three are compared under the default. Same tasks, slots, metrics and statistics; this brief refreshes as runs finish.

### What weight-space programming did on each task, on unseen starts

| | **Cosmos3 Edge Policy**<br>world-action model | **GR00T N1.7**<br>reasoning-backbone VLA | **π0.5**<br>OpenPI VLA, control · earlier slots |
|---|---|---|---|
| T1 Cube then banana | *running* | unedited 9 of 32<br>not edited yet | fixed: 27 → 106 of 128 earlier slots<br>distilled edit; 81 / 2, $p = 7.2 \times 10^{-22}$ |
| T2 Pick up the eyeglasses | *running* | *running* | T1 edit: 22 → 44 of 64 earlier slots<br>own edit 18 → 27 of 64, not significant |
| T3 Cube left of the bowl | *running* | *running* | T1 edit: 8 → 36 of 64 earlier slots<br>no own edit submitted |
| T4 Stack yellow on red | *running* | *running* | no gain earlier slots<br>lookalike perception; no clear wording |
| T5 Mustard into the left bin | *running* | *running* | no gain earlier slots<br>own edit 8 → 11 of 64; the grasp slips |
| T6 Stack three cubes | *running* | *running* | no gain earlier slots<br>own edit 35 → 39 of 64 |
| T7 Pick up the flat box | *running* | *running* | no gain earlier slots<br>a missing skill; 0 of 64 unedited |
| T8 Mustard right of the raisin box, reference-first | unedited 0 of 16<br>not edited yet | *running* | fixed: 0 → 27 of 64 earlier slots<br>distilled edit; 27 / 0, $p = 1.5 \times 10^{-8}$ |
| R1 Retention task | *running* | *running* | kept: 32 → 32 of 32 earlier slots<br>cube into the bowl, T1 edit |
| Rendering: performance → default preset | *running* | 5 → 16 of 16<br>Banana onto plate | *running* |
| Random control, same size | *running* | *running* | 9 of 64 earlier slots<br>T1; unedited 13, edit 50 |
| Where to edit | *running* | *running* | VLM 55, action expert 20, head 0 of 64 earlier slots<br>T1; unedited 5; only the VLM edit transfers |

| | **Cosmos3 Edge** | **GR00T N1.7** | **π0.5** |
|---|---|---|---|
| T1 Cube then banana | *running* | 9/32 · 28% | 27/128 · 21% |
| T2 Eyeglasses | *running* | *running* | 18/64 · 28% |
| T3 Cube left of bowl | *running* | *running* | 8/64 · 13% |
| T4 Stack yellow on red | *running* | *running* | 13/32 · 41% |
| T5 Mustard into left bin | *running* | *running* | 8/64 · 13% |
| T6 Stack three cubes | *running* | *running* | 35/64 · 55% |
| T7 Flat box | *running* | *running* | 0/32 · 0% |
| T8 Mustard, reference-first | 0/16 · 0% | *running* | 0/64 · 0% |
| R1 Retention task | *running* | *running* | 32/32 · 100% |

Unedited success on each shared task; stronger shading is higher. Each cell gives successes of starts.

| | **Cosmos3 Edge** | **GR00T N1.7** | **π0.5** |
|---|---|---|---|
| T1 Cube then banana | *running* | *running* | **+79 / 128*** |
| T2 Eyeglasses | *running* | *running* | **+22 / 64*** |
| T3 Cube left of bowl | *running* | *running* | **+28 / 64*** |
| T4 Stack yellow on red | *running* | *running* | +3 / 32 |
| T5 Mustard into left bin | *running* | *running* | +5 / 32 |
| T6 Stack three cubes | *running* | *running* | +5 / 32 |
| T7 Flat box | *running* | *running* | ±0 / 32 |
| T8 Mustard, reference-first | *running* | *running* | not run |
| R1 Retention task | *running* | *running* | ±0 / 32 |

gain, $p < 0.05$ loss, $p < 0.05$ no detectable change

* $p < 0.05$; outlined: the edit's own task

Each model's main edit, unchanged, on every suite task: edited minus unedited successes on the same starts. Teal: a gain, dark: a loss (exact McNemar $p < 0.05$); grey: no detectable change; the outlined cell is the task the edit was built for.

Live page with every table, figure and record: ali-adeeb.com/weightsmith/cosmos · Paper: ali-adeeb.com/weightsmith/paper.pdf · Code and records: github.com/adeeb10abbas/weightsmith

Draft for discussion. Simulation only (RoboLab, Isaac Sim), one released checkpoint per model, shared slots 30000–30031 and 96000–96063 with identical initial scenes; π0.5 cells are earlier records on their own slots until its shared-slot runs land. Exact McNemar tests on paired starts.